

Foundation (Jalapeno)

- Q1.** What is the purpose of a two-way frequency table in sports statistics?
- (A) To compare team scores
 - (B) To analyze player performance
 - (C) To identify associations between variables
 - (D) To predict game outcomes
 - (E) To calculate relative frequencies
- Q2.** A basketball team won 8 games out of 12 when their best player was on the court. What is the relative frequency of winning with the best player?
- (A) $\frac{1}{3}$
 - (B) $\frac{2}{3}$
 - (C) $\frac{1}{2}$
 - (D) $\frac{3}{4}$
 - (E) $\frac{1}{4}$
- Q3.** A two-way table shows the number of athletes who participated in a marathon, categorized by age group and finishing status. What can be inferred from the table?
- (A) The fastest finishing time
 - (B) The average age of participants
 - (C) The association between age group and finishing status
 - (D) The total number of participants
 - (E) The winning time
- Q4.** A sports analyst created a two-way table to compare the performance of two tennis players, *A* and *B*, based on the surface they played on. Which of the following can be calculated from the table?
- (A) The number of games won by player *A* on clay courts
 - (B) The relative frequency of player *B* winning on hard courts
 - (C) The association between player performance and surface type
 - (D) The total number of games played by both players
 - (E) The probability of player *A* winning on grass courts
- Q5.** A cycling team recorded the distance covered by their riders, categorized by the type of bike used. Which of the following is an example of a two-way table?
- (A) Distance covered vs. rider experience
 - (B) Type of bike vs. distance covered
 - (C) Rider speed vs. distance covered
 - (D) Type of bike vs. rider experience
 - (E) Distance covered vs. rider speed
- Q6.** A two-way table shows the number of soccer players who scored goals, categorized by their position on the field. What can be calculated from the table?
- (A) The total number of goals scored
 - (B) The relative frequency of goals scored by each position
 - (C) The association between position and goal-scoring ability
 - (D) The average number of goals scored per position
 - (E) The probability of a player scoring a goal
- Q7.** A swimmer competed in 15 events, winning 9 of them. What is the relative frequency of winning?
- (A) $\frac{1}{5}$
 - (B) $\frac{3}{5}$
 - (C) $\frac{2}{3}$
 - (D) $\frac{3}{4}$
 - (E) $\frac{1}{3}$
- Q8.** A two-way table shows the number of athletes who participated in a triathlon, categorized by their age group and finishing status. What is the purpose of calculating relative frequencies from the table?
- (A) To compare the number of participants in each age group
 - (B) To identify the most common finishing status
 - (C) To analyze the association between age group and finishing status
 - (D) To calculate the total number of participants
 - (E) To determine the winning time

Open Ended Questions (FRQ)

- Q9.** Create a two-way table to compare the performance of two runners, *X* and *Y*, based on the distance they ran. Calculate the relative frequencies and analyze the association between runner performance and distance.
- Q10.** A volleyball team recorded the number of points scored, categorized by the type of serve used. Create a two-way table and calculate the relative frequencies to analyze the association between serve type and points scored.

Chef's Tips

- Tip 1: Ensure students understand the concept of two-way tables before administering the test.
- Tip 2: Remind students to read each question carefully and choose the correct answer.
- Tip 3: Allow students to use calculators to calculate relative frequencies.